# Supplementary Material for: Data-Driven Nonlinear Output Regulation with PAC Guarantees

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## I. Introduction

This supplement gives the numerical objects and intermediate results used in Section VIII of *Data-Driven Nonlinear Output Regulation with PAC Guarantees*. Section II specifies the system data, finite sets, contraction experiments, RKHS configuration, and validation parameters used in Section VIII-A of the main paper. Section III reports the initial comparison and each application of (56) and (57) used in Section VIII-B of the main paper. Section IV gives the quantities used to evaluate (47) in Section VIII-C of the main paper.

## II. Numerical Settings

The coefficients in Table I and the relation

$$c_{\ell,-k_1,-k_2} = \overline{c_{\ell,k_1,k_2}}$$

specify $q_r$ in Section VIII-A.

TABLE I: The independent coefficients specify the reference in Section VIII-A.

| $\ell$ | $k_1$ | $k_2$ | $\mathrm{Re}\, c_{\ell,k_1,k_2}$ | $\mathrm{Im}\, c_{\ell,k_1,k_2}$ |
|---|---|---|---|---|
| 1 | 0 | 0 | 5.91821e-08 | 0 |
| 1 | 0 | 1 | -1.93593e-05 | -1.21575e-04 |
| 1 | 0 | 2 | 5.70836e-09 | 5.09526e-09 |
| 1 | 0 | 3 | -6.56615e-11 | -4.40107e-12 |
| 1 | 1 | 0 | 0.00120601 | 7.20684e-04 |
| 1 | 1 | 1 | 1.53334e-04 | -3.53327e-05 |
| 1 | 1 | 2 | -5.99641e-07 | 4.24072e-07 |
| 1 | 1 | 3 | 1.70138e-09 | -3.37094e-09 |
| 1 | 1 | -3 | 6.66674e-11 | 2.23195e-10 |
| 1 | 1 | -2 | 2.34533e-08 | 3.42647e-08 |
| 1 | 1 | -1 | 5.13039e-06 | 4.98278e-06 |
| 1 | 2 | 0 | -1.00445e-10 | 2.10195e-08 |
| 1 | 2 | 1 | -4.03550e-07 | -6.20256e-07 |
| 1 | 2 | 2 | 1.35813e-08 | 7.52644e-09 |
| 1 | 2 | 3 | -2.11663e-10 | 1.61021e-11 |
| 1 | 2 | -3 | -1.43642e-11 | 4.91295e-12 |
| 1 | 2 | -2 | -7.85883e-10 | 5.84243e-10 |
| 1 | 2 | -1 | -2.79304e-08 | 3.14822e-08 |
| 1 | 3 | 0 | 2.61163e-11 | -8.62894e-10 |
| 1 | 3 | 1 | -2.33181e-09 | 3.05355e-09 |
| 1 | 3 | 2 | -3.43073e-11 | -2.22026e-10 |
| 1 | 3 | 3 | 3.13102e-12 | 5.67410e-12 |
| 1 | 3 | -2 | -1.07295e-11 | -1.16455e-11 |
| 1 | 3 | -1 | -2.03877e-10 | -1.47009e-10 |
| 2 | 0 | 0 | -6.89225e-08 | 0 |
| 2 | 0 | 1 | -3.23609e-05 | -2.44572e-04 |
| 2 | 0 | 2 | 1.02348e-08 | 7.86945e-09 |
| 2 | 0 | 3 | -1.08234e-10 | -1.00299e-11 |
| 2 | 1 | 0 | 0.00146439 | 0.00149113 |

| $\ell$ | $k_1$ | $k_2$ | $\operatorname{Re} c_{\ell,k_1,k_2}$ | $\operatorname{Im} c_{\ell,k_1,k_2}$ |
|---|---|---|---|---|
| 2 | 1 | 1 | 1.74595e-04 | -5.54715e-05 |
| 2 | 1 | 2 | -9.19565e-07 | 6.86640e-07 |
| 2 | 1 | 3 | 2.72658e-09 | -5.48146e-09 |
| 2 | 1 | -3 | 1.08949e-10 | 3.62952e-10 |
| 2 | 1 | -2 | 3.79277e-08 | 5.59387e-08 |
| 2 | 1 | -1 | 7.95076e-06 | 8.34075e-06 |
| 2 | 2 | 0 | -1.62707e-08 | -4.20514e-08 |
| 2 | 2 | 1 | -6.66966e-07 | -9.58366e-07 |
| 2 | 2 | 2 | 2.15867e-08 | 1.26108e-08 |
| 2 | 2 | 3 | -3.37478e-10 | 1.90215e-11 |
| 2 | 2 | -3 | -2.32097e-11 | 7.86909e-12 |
| 2 | 2 | -2 | -1.26959e-09 | 9.36752e-10 |
| 2 | 2 | -1 | -4.51096e-08 | 4.99938e-08 |
| 2 | 3 | 0 | 2.18229e-09 | -1.99371e-09 |
| 2 | 3 | 1 | -3.64324e-09 | 4.67874e-09 |
| 2 | 3 | 2 | -5.93337e-11 | -3.62958e-10 |
| 2 | 3 | 3 | 5.56743e-12 | 9.52966e-12 |
| 2 | 3 | -2 | -1.67583e-11 | -1.90109e-11 |
| 2 | 3 | -1 | -2.92052e-10 | -2.53214e-10 |

For $X_a^{(L)} = [-L, L]^6$, $L \in \{5, 8, 10\}$, the following matrix satisfies (15):

$$R_Q^{(L)} = \begin{bmatrix} 0.1\sqrt{2}L & 0 & \sqrt{3} & 0 & 0 \\ 0 & 0.1\sqrt{2}L & 0 & \sqrt{3} & 0 \\ 0 & 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The first two entries bound $\nabla Q_x(x)$ on the selected $X_a^{(L)}$. The remaining nonzero entries use the derivative bound of the output map in Assumption 2.

The four elements of $\mathcal{W}_{\mathrm{SDP}}$, in the order $(\varpi_\alpha, \varpi_P, \varpi_Y, \varpi_G)$, are

$$(0.9997, 0, 0.00015, 0.00015) \qquad (0.9998, 0.0001, 0.00005, 0.00005)$$
$$(0.9995, 0.0001, 0.0002, 0.0002) \quad (0.9999, 0, 0.00005, 0.00005).$$

The finite sets used by Algorithm 1 are

TABLE II: The finite sets specify the initial candidates and the allowed updates.

| | set | value |
|---|---|---|
| 1 | $\{\chi_j\}$ | $\chi_j(w) = 1$ for $j = 1, 2, 3$ |
| 2 | $\{\Psi_j\}$ | $\Psi_j = 1$ for $j = 1, 2, 3$ |
| 3 | $\{\zeta_{j,b_j}\}$ | no nonlinear readout feature, with $d_{j,b_j} = 0$ |
| 4 | $\{c_{j,b_j}\}$ | no nonlinear readout coefficient |
| 5 | $\mathcal{I}_{G,j}$ | the singleton set containing the pair $(a_j, b_j)$ for which $\Phi_{g,j}^{(a_j)} = 0$, $\Psi_j^{(a_j)} = 1$, and $\beta_{g,j}^{(a_j,b_j)}(\vartheta_j) = \vartheta_j$ |
| 6 | $\mathcal{I}_{R,j}(a_j)$ | the singleton set containing the index $r_j$ for which $(M_j^{(a_j,r_j)}, N_j^{(a_j,r_j)}) = (-10, 1/2)$ |
| 7 | $\mathcal{G}_\Gamma(3)$ | $\{\mathrm{col}(0.2I_2, 0_{1\times 2}), \mathrm{col}(0.6I_2, 0_{1\times 2}), \mathrm{col}(I_2, 0_{1\times 2})\}$ |
| 8 | $\mathcal{I}_{\mathcal{A}}$ | the three dimension-compatible tuples $\big((a_j, b_j, r_j)_{j=1}^3, \Gamma\big)$ satisfying $(a_j, b_j) \in \mathcal{I}_{G,j}$, $r_j \in \mathcal{I}_{R,j}(a_j)$, and $\Gamma \in \mathcal{G}_\Gamma(3)$ |
| 9 | $\mathcal{W}_{\mathrm{SDP}}$ | the four rows displayed above |
| 10 | initial RKHS set | the Fourier kernel with $\lvert\Xi_0\rvert = 16$, $b_\kappa = 1$, and $W_v = I_6$ |
| 11 | $\mathcal{D}_{j,q}$ | empty |
| 12 | $\mathcal{X}_{\mathrm{cen}}$ | the 121 exosystem states defined below |

| | set | value |
|---|---|---|
| 13 | $\mathcal{W}_{v,q}^{+}$ | $\{W_{v,q}, 10I_6\}$ |

For $\theta = (\theta_1, \theta_2)$, define $\phi(w) = \mathrm{col}(1, \{\cos(k^\top\theta), \sin(k^\top\theta)\}_{k\in\mathcal{K}_+}) \in \mathbb{R}^{49}$. The 121 elements of $\mathcal{X}_{\mathrm{cen}}$ satisfy

$$\theta(\xi_n) = 2\pi \bmod \left( \begin{bmatrix} 0.137 \\ 0.383 \end{bmatrix} + n \begin{bmatrix} (\sqrt{5}-1)/2 \\ \sqrt{2}-1 \end{bmatrix}, 1 \right), \qquad n = 0, \ldots, 120.$$

Column pivoted QR factorization of $[\phi(\xi_0) \ \cdots \ \phi(\xi_{120})]$ orders the exosystem states by the independent directions that their feature columns add. The first 16 points form $\Xi_0$. Fig. 1(a) shows the $24^2$ points in $\mathcal{D}_{\mathrm{gen}}$. Fig. 1(b) shows the $64^2$ points in $\mathcal{N}_{h_W}$. Fig. 1(c) shows the 121 points in $\mathcal{X}_{\mathrm{cen}}$ and the 16 selected centers in $\Xi_0$.

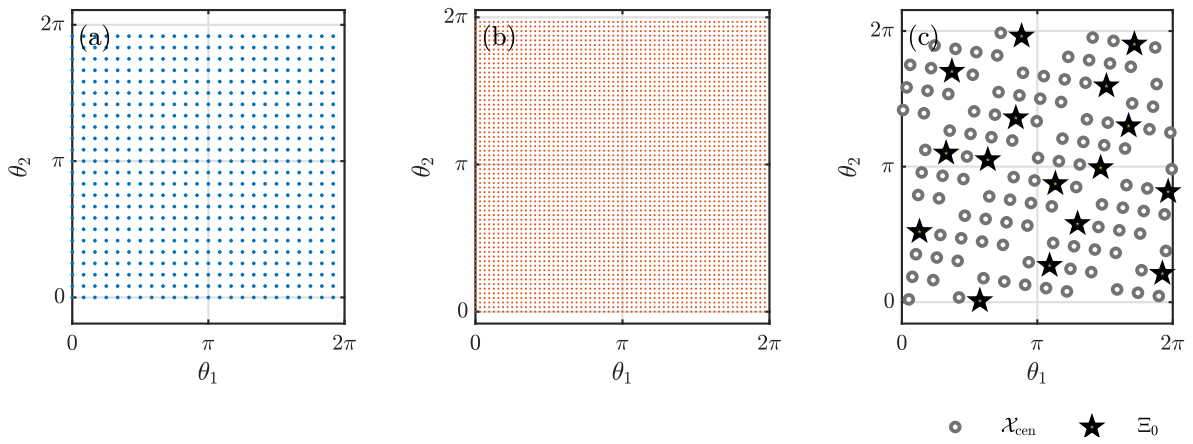



Fig. 1. The prescribed exosystem states form the sets used in the numerical example. (a) The $24^2$ points in $\mathcal{D}_{\mathrm{gen}}$. (b) The $64^2$ points in $\mathcal{N}_{h_W}$. (c) The 121 points in $\mathcal{X}_{\mathrm{cen}}$, with the 16 points in $\Xi_0$ marked by stars.

The numerical objects outside these finite sets are

TABLE III: Numerical settings used in the example.

| | object | value |
|---|---|---|
| 1 | $\lambda_\Phi$ | $10^{-10}$ |
| 2 | $\bar{\mathcal{E}}_{\mathrm{clos},j}$ | $10^{-12}$ for $j = 1, 2, 3$ |
| 3 | $X_a$ | $X_a^{(L)}$ for $L \in \{5, 8, 10\}$ |
| 4 | $R_Q$ | the expression displayed above for $L \in \{5, 8, 10\}$ |
| 5 | $\Delta_d$ | $\sqrt{16}\bar{d}I_2$ |
| 6 | $\mathcal{D}_{\mathrm{con}}^{\mathrm{raw}}$ | the 16 records in Tables V and VI |
| 7 | initial compensator states | the 16 columns displayed after Tables V and VI |
| 8 | $\Delta_R$ | unused because $r_X(c)$ is finite |
| 9 | $\mathcal{O}_W$ | an open neighborhood of $W$ used for $\bar{D}_\phi$ |
| 10 | $h_W$ | 0.06942 |
| 11 | $\epsilon_{\mathrm{safe}}$ | $10^{-4}$ |
| 12 | $W_v$ | $I_6$ initially and $10I_6$ after the selected $U_{\mathrm{inv}}$ update |
| 13 | $W_z$ | $I_6$ |
| 14 | $\rho_z$ | 1 |
| 15 | $\rho_\delta$ | 1 |
| 16 | $\nu$ | $10^{-8}$ |
| 17 | $\bar{D}_\phi$ | 191.47 |
| 18 | $\bar{s}$ | 1.900 |
| 19 | $\mathcal{D}_{\mathrm{fit},q}$ | 96 terminal records with $T_{\mathrm{fit}} = 30$ and integration step 0.004 |
| 20 | $T_{\mathrm{qry}}$ | 30 |
| 21 | $\mathcal{D}_{\mathrm{qry}}$ | terminal records generated from $c_q$ and the backward-flow exosystem state in Section V |
| 22 | $\mu_{\mathrm{tune}}$ | $\theta_1$ and $\theta_2$ are independent and uniform on $[0, 2\pi)$, with $w = (\cos\theta_1, \sin\theta_1, \cos\theta_2, \sin\theta_2)$ |

| | object | value |
|---|---|---|
| 23 | $N_{\text{tune}}$ | 64 |
| 24 | $M_{\text{tune}}$ | $2 \times 10^{15}$ |
| 25 | $\beta_{\text{tune}}$ | 0.02 |
| 26 | $\gamma_{\text{tune}}$ | 0.1 |
| 27 | $\rho_{\text{safe}}$ | 0.99 |
| 28 | $(b_{\text{out}}, b_{\text{inv}}, b_{\text{graph}}, b_{\text{MC}})$ | $(0.85, 0.02, 0.02, 0.11)$ |
| 29 | $\epsilon_g$ | $10^{-8}$ |
| 30 | $\epsilon_u$ | $10^{-8}$ |
| 31 | $\mathcal{J}_{\Phi}^{(q)}$ | empty |
| 32 | $\mathcal{J}_{\beta}^{(q)}$ | empty |
| 33 | $d_W(w, \xi)$ | $\|w - \xi\|_2$ |
| 34 | $N_{\text{add}}$ | 17 at $q = 0$ and 16 thereafter |
| 35 | $\rho_{\text{hard}}$ | 1.0813 |
| 36 | $\delta_D$ | 0.01 |
| 37 | $\delta_\chi$ | 0.05 for all $\chi \in \mathcal{I}_{\text{comp}}$ |
| 38 | $Q_{\max}$ | 5 |
| 39 | $(c_\eta, c_Z, c_\kappa, c_\beta)$ | $(0, 0, 1, 0)$ |
| 40 | $(n_{\eta,\max}, s_{a,\max}, N_{\kappa,\max}, d_{\beta,\max})$ | $(1, 1, 121, 1)$ |

The raw contraction data use one zero input record and 15 records obtained with

$$p_{\text{PE},j}(t) = 0.3 \sum_{\ell=1}^{15} a_{\ell j} \sin(\omega_\ell t + \phi_\ell), \qquad j = 1, 2, 3.$$

Fig. 2(a), Fig. 2(b), and Fig. 2(c) show $u_1(t)$, $u_2(t)$, and $u_3(t)$, respectively. The circles in each part mark the 15 sampling instants used to form the records obtained with the prescribed excitation input in $\mathcal{D}_{\text{con}}^{\text{raw}}$. The remaining record uses the prescribed zero input. Table IV gives all nonzero coefficients. Tables V and VI give every tuple $(x_i, y_i, u_i, r_i, \dot{x}_i)$ in $\mathcal{D}_{\text{con}}^{\text{raw}}$, with $r_i = q_r(w_i)$.

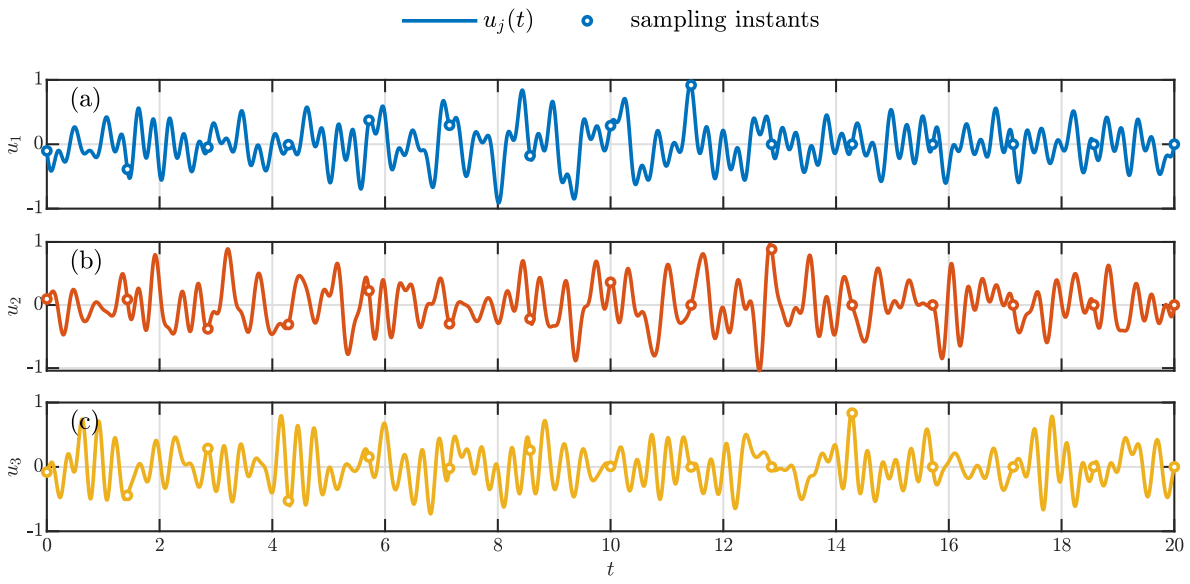



Fig. 2. The prescribed excitation input generates $\mathcal{D}_{\text{con}}^{\text{raw}}$. (a) $u_1(t)$. (b) $u_2(t)$. (c) $u_3(t)$. In each part, the circles mark the 15 sampling instants used to form these records.

TABLE IV: The nonzero coefficients define the excitation used for $\mathcal{D}_{\text{con}}^{\text{raw}}$.

| $\ell$ | $\omega_\ell$ | $\phi_\ell$ | $a_{\ell 1}$ | $a_{\ell 2}$ | $a_{\ell 3}$ |
|---|---|---|---|---|---|
| 1 | 3.72713 | 2.15826 | 0.238626 | 0.529455 | -0.398422 |

| $\ell$ | $\omega_\ell$ | $\phi_\ell$ | $a_{\ell 1}$ | $a_{\ell 2}$ | $a_{\ell 3}$ |
|---|---|---|---|---|---|
| 2 | 4.3202 | 2.27764 | 0.424404 | 0.117832 | 0.127201 |
| 3 | 4.73938 | 4.49902 | 0.593143 | 0.324967 | 0.382413 |
| 4 | 9.65187 | 2.52985 | 0.347932 | 0.732326 | -0.0657024 |
| 5 | 10.0634 | 1.59854 | 0.287779 | -0.365019 | 0.0213582 |
| 6 | 10.1605 | 1.2775 | -0.812699 | 0.355859 | 0.1311 |
| 7 | 10.464 | 2.20024 | 0.461747 | -0.575025 | 0.420277 |
| 8 | 13.7099 | 4.05857 | 0.0966734 | -0.579094 | -0.192694 |
| 9 | 15.7251 | 4.52874 | 0.142717 | -0.261071 | -0.135434 |
| 10 | 17.4083 | 1.76489 | -0.156342 | -0.497643 | -0.486576 |
| 11 | 19.7574 | 5.15172 | -0.14233 | 0.250134 | -0.647521 |
| 12 | 20.0867 | 3.59887 | 0.00421145 | -0.363464 | -0.301791 |
| 13 | 21.1807 | 4.94975 | 0.406087 | 0.0353736 | -0.18634 |
| 14 | 23.0672 | 2.23756 | 0.686501 | 0.478414 | -0.582365 |
| 15 | 24.863 | 4.25714 | 0.299049 | 0.367175 | 0.399872 |

TABLE V: States and inputs in the raw contraction records.

| $i$ | $x_1$ | $x_2$ | $x_3$ | $u_1$ | $u_2$ | $u_3$ |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | -1.98996 | -1.99891 | -0.462068 | -0.102879 | 0.0954457 | -0.0802551 |
| 3 | 1.91264 | 1.94477 | -0.366037 | -0.388862 | 0.0880935 | -0.443111 |
| 4 | -1.94221 | 1.89074 | -0.123442 | -0.0461247 | -0.378217 | 0.285792 |
| 5 | 1.26205 | -1.70537 | 0.822681 | -0.0071387 | -0.307799 | -0.525702 |
| 6 | 1.48932 | -1.6415 | -0.526679 | 0.373835 | 0.225909 | 0.15441 |
| 7 | -0.941274 | 0.0189941 | 0.931044 | 0.295882 | -0.29739 | -0.0225332 |
| 8 | 0.310293 | 0.282946 | 0.555453 | -0.177435 | -0.218764 | 0.260687 |
| 9 | -0.732417 | 1.91803 | 0.95957 | 0.289055 | 0.360337 | 0.00822697 |
| 10 | 0.486611 | 0.79416 | -0.486753 | 0.918525 | 0 | 0 |
| 11 | -0.690662 | -0.0488545 | 0.515214 | 0 | 0.882137 | 0 |
| 12 | 1.89315 | -0.68428 | 0.261272 | 0 | 0 | 0.834744 |
| 13 | -0.690049 | 1.73386 | 0.740624 | 0 | 0 | 0 |
| 14 | -1.66476 | 0.99092 | 0.661881 | 0 | 0 | 0 |
| 15 | 1.54381 | 1.74922 | 0.578117 | 0 | 0 | 0 |
| 16 | -0.0905904 | 0.115725 | -0.512863 | 0 | 0 | 0 |

TABLE VI: References, outputs, and state derivatives in the raw contraction records.

| $i$ | $q_{r,1}$ | $q_{r,2}$ | $y_1$ | $y_2$ | $\dot{x}_1$ | $\dot{x}_2$ | $\dot{x}_3$ |
|---|---|---|---|---|---|---|---|
| 1 | 0.00233111 | 0.0029428 | 0 | 0 | 0.0441022 | 0.00234145 | 0 |
| 2 | 0.00182671 | 0.00166045 | -0.985276 | -0.985535 | 131.222 | -91.3108 | 5.01846 |
| 3 | 3.25848e-04 | -6.69076e-04 | 0.913224 | 0.918403 | 79.4736 | -118.357 | 4.03239 |
| 4 | -5.56042e-04 | -0.00160904 | -0.968384 | 0.943313 | 127.331 | -127.271 | 1.60177 |
| 5 | -0.00152031 | -0.00324666 | 0.96955 | -0.707766 | 43.5956 | -39.5302 | -8.00176 |
| 6 | -0.00223296 | -0.00379804 | 0.745454 | -0.97417 | 57.1197 | -55.0615 | 5.51242 |
| 7 | -0.00296228 | -0.00430856 | -0.0102296 | 0.7398 | 21.8803 | -12.733 | -9.26612 |
| 8 | -0.00274017 | -0.00390806 | 0.699206 | 0.68496 | 2.85718 | -9.2526 | -5.54572 |
| 9 | -0.00200546 | -0.00222534 | 0.223326 | 0.993687 | 67.2966 | -78.5402 | -9.38494 |
| 10 | -0.00198302 | -0.00193639 | -1.42378e-04 | 0.298076 | 8.26788 | -20.1906 | 4.91091 |
| 11 | -4.25848e-04 | 3.58009e-04 | -0.17367 | 0.435253 | 13.6554 | -5.37477 | -5.12817 |
| 12 | 9.22490e-04 | 0.0024133 | 0.973459 | -0.399461 | 50.1701 | -62.2161 | -2.41011 |
| 13 | 0.00102802 | 0.00230611 | 0.050532 | 0.985918 | 56.1452 | -66.5449 | -7.23212 |
| 14 | 0.00257542 | 0.00436803 | -0.762801 | 0.929241 | 69.7502 | -62.9408 | -6.43115 |
| 15 | 0.00301914 | 0.00448693 | 0.971702 | 0.981146 | 61.5831 | -94.4346 | -5.50901 |
| 16 | 0.00227359 | 0.00301939 | -0.539503 | -0.377498 | 1.25243 | -1.45224 | 5.12971 |

For the $i$th raw contraction experiment, the initial compensator state is the $i$th column of the following matrix.

$$\begin{bmatrix} 0 & -0.9886 & -0.8754 & 0.7786 & 0.5832 & 0.6727 & -0.4570 & 0.5358 & -0.7596 & -0.2217 & 0.8486 & -0.7556 & -0.3392 & -0.7861 & -0.6029 & 0.0006 \\ 0 & -0.4658 & 0.4045 & -0.1267 & -0.9960 & 0.7251 & 0.9591 & -0.9794 & 0.2503 & -0.7670 & -0.5993 & -0.2188 & -0.7835 & -0.5878 & 0.3325 & -0.9842 \\ 0 & 0.7890 & 0.4640 & 0.7758 & -0.4540 & 0.2276 & 0.3765 & -0.2291 & 0.7705 & 0.1049 & -0.0941 & -0.0451 & -0.2923 & 0.4288 & -0.9306 & -0.5137 \end{bmatrix}.$$

For each compiled architecture, the compensator in (9) is replayed from these initial states on the fixed raw data. The resulting values of $\eta_i$ and $\dot{\eta}_i$ are used with $\mathcal{D}_{\mathrm{con}}^{\mathrm{raw}}$ to construct $U_0$, $Z_0$, and $Z_1$. These matrices satisfy (16) and (20) for the architecture under evaluation.

## III. Residual Budget Selection

Table VII reports all 36 initial cases.

TABLE VII: The initial finite set is screened by the contraction tube and RKHS conditions.

| $g$ | $L$ | $\varpi$ | $\lambda$ | $\operatorname{rank} Z_0$ | (17) | $\mathcal{T}$ | (36) | retained |
|---|---|---|---|---|---|---|---|---|
| 0.2 | 5 | 1 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.2 | 5 | 2 | 1.94831 | 9 | satisfied | available | solved | yes |
| 0.2 | 5 | 3 | 1.79094 | 9 | satisfied | available | solved | yes |
| 0.2 | 5 | 4 | 1.91325 | 9 | satisfied | available | solved | yes |
| 0.6 | 5 | 1 | 1.82584 | 9 | satisfied | available | solved | yes |
| 0.6 | 5 | 2 | 1.94736 | 9 | satisfied | available | solved | yes |
| 0.6 | 5 | 3 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.6 | 5 | 4 | 1.9192 | 9 | satisfied | available | solved | yes |
| 1 | 5 | 1 | 1.82581 | 9 | satisfied | available | solved | yes |
| 1 | 5 | 2 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 1 | 5 | 3 | 1.79054 | 9 | satisfied | available | solved | yes |
| 1 | 5 | 4 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.2 | 8 | 1 | 1.81937 | 9 | satisfied | available | solved | yes |
| 0.2 | 8 | 2 | 1.94589 | 9 | satisfied | available | not evaluable | no |
| 0.2 | 8 | 3 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.2 | 8 | 4 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.6 | 8 | 1 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.6 | 8 | 2 | 1.94545 | 9 | satisfied | available | not evaluable | no |
| 0.6 | 8 | 3 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.6 | 8 | 4 | 1.90286 | 9 | satisfied | available | solved | yes |
| 1 | 8 | 1 | 1.8207 | 9 | satisfied | available | solved | yes |
| 1 | 8 | 2 | 1.94592 | 9 | satisfied | unavailable | not evaluated | no |
| 1 | 8 | 3 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 1 | 8 | 4 | 1.8849 | 9 | satisfied | available | solved | yes |
| 0.2 | 10 | 1 | 1.81754 | 9 | satisfied | available | solved | yes |
| 0.2 | 10 | 2 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.2 | 10 | 3 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.2 | 10 | 4 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.6 | 10 | 1 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 0.6 | 10 | 2 | 1.94036 | 9 | satisfied | available | not evaluable | no |
| 0.6 | 10 | 3 | 1.7798 | 9 | satisfied | available | solved | yes |
| 0.6 | 10 | 4 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 1 | 10 | 1 | – | 9 | not satisfied | unavailable | not evaluated | no |
| 1 | 10 | 2 | 1.9395 | 9 | satisfied | available | not evaluable | no |
| 1 | 10 | 3 | 1.77958 | 9 | satisfied | available | solved | yes |
| 1 | 10 | 4 | – | 9 | not satisfied | unavailable | not evaluated | no |

"Not evaluated" means that (36) was not solved because (17) or (22) was not satisfied. "Not evaluable" means that $\mathcal{D}_{\mathrm{fit}}$ or $\mathcal{D}_{\mathrm{qry}}(\Omega)$ could not be obtained from the corresponding trajectories, so (36) and the residuals were not evaluated.

The 15 retained cases are compared through the ordered triple $(D_{\max}, C_{\mathrm{tail}}, \mathrm{Comp})$ in (57). Table VIII shows this comparison and its first row in the ordering defines $\mathcal{C}_0$.

TABLE VIII: The retained initial candidates are compared by (57).

| $g$ | $L$ | $\varpi$ | $D_{\mathrm{out}}$ | $D_{\mathrm{inv}}$ | $D_{\mathrm{graph}}$ | $D_{\mathrm{MC}}$ | $D_{\max}$ | $C_{\mathrm{tail}}$ | Comp | rank |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.2 | 5 | 4 | 1.06904 | 137.318 | 1.32109e+04 | 0.637256 | 1.32109e+04 | 1.0585 | 0.132231 | 1 |
| 1 | 8 | 4 | 1.11601 | 146.319 | 1.48503e+04 | 1.08753 | 1.48503e+04 | 1.18934 | 0.132231 | 2 |
| 0.6 | 5 | 4 | 1.04925 | 151.453 | 1.59280e+04 | 0.740788 | 1.59280e+04 | 1.27458 | 0.132231 | 3 |
| 1 | 5 | 1 | 0.924399 | 189.093 | 1.92714e+04 | 0.836883 | 1.92714e+04 | 1.54155 | 0.132231 | 4 |
| 0.6 | 5 | 1 | 0.941324 | 186.251 | 1.93378e+04 | 0.869305 | 1.93378e+04 | 1.54668 | 0.132231 | 5 |
| 0.6 | 8 | 4 | 1.12573 | 162.577 | 2.09246e+04 | 1.62313 | 2.09246e+04 | 1.67169 | 0.132231 | 6 |
| 0.2 | 5 | 3 | 0.932616 | 185.8 | 2.10036e+04 | 0.970674 | 2.10036e+04 | 1.6786 | 0.132231 | 7 |
| 1 | 5 | 3 | 0.900191 | 200.276 | 2.36156e+04 | 1.02671 | 2.36156e+04 | 1.88636 | 0.132231 | 8 |
| 0.2 | 8 | 1 | 1.04435 | 211.572 | 2.89917e+04 | 2.32461 | 2.89917e+04 | 2.31386 | 0.132231 | 9 |
| 1 | 8 | 1 | 1.00289 | 242.426 | 3.66282e+04 | 2.72205 | 3.66282e+04 | 2.92071 | 0.132231 | 10 |
| 1 | 10 | 3 | 1.03975 | 278.662 | 5.26050e+04 | 5.10245 | 5.26050e+04 | 4.1896 | 0.132231 | 11 |
| 0.6 | 10 | 3 | 1.05871 | 282.077 | 5.88147e+04 | 5.97923 | 5.88147e+04 | 4.68205 | 0.132231 | 12 |
| 0.2 | 10 | 1 | 1.10247 | 339.635 | 1.15670e+05 | 12.5731 | 1.15670e+05 | 9.1917 | 0.132231 | 13 |
| 0.6 | 5 | 2 | 1.03508 | 321.596 | 1.27214e+05 | 6.31953 | 1.27214e+05 | 10.1016 | 0.132231 | 14 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.2 | 5 | 2 | 1.05898 | 341.911 | 1.66560e+05 | 8.72193 | 1.66560e+05 | 13.2202 | 0.132231 | 15 |

The common tuning panel at $q = 0$ uses $M_{\mathrm{tune}} = 2 \times 10^{15}$ for the 64 primary exosystem states. The four terms are those in (53), and their empirical tail quantities are defined immediately after (54). For each primary exosystem state, the $M_{\mathrm{tune}}$ independent unit exponential variables are represented by one multinomial count vector on the 8193 intervals used in Remark 8 of the main paper. The 64 count vectors therefore represent $64M_{\mathrm{tune}} = 1.28 \times 10^{17}$ unit exponential variables. Their entries are used in the sums defining $c^{(q)}_{\mathrm{inv,i}}$ and $c^{(q)}_{\mathrm{graph,i}}$ in (53), and all four terms in (53) are recomputed with $M_{\mathrm{tune}} = 2 \times 10^{15}$.

For each $q$, the following tables first display $\mathcal{U}_q$ and then test $\mathcal{U}_q^{\mathrm{eval}}$ by (56). All numerical entries in the following comparison tables are rounded for presentation, while the acceptance tests and lexicographic rankings use the unrounded values. For compact presentation, write $\Gamma(g) := \mathrm{col}(gI_2, 0_{1\times 2})$, where $g \in \{0.2, 0.6, 1\}$. In the proposal tables, the update quantities are $(\Gamma^+, \varpi)$ for $U_{\mathrm{out}}$, $(|\Xi^+|, b^+_\kappa)$ for $U_{\mathrm{graph}}$, and $(|\Xi^+|, \lambda_{\min}(W^+_v))$ for $U_{\mathrm{inv}}$. At $q = 0$, the required bounds are $D_{\max}(\mathcal{C}) \le 0.99 D_{\max}(\mathcal{C}_0)$, $C_{\mathrm{tail}}(\mathcal{C}) \le C_{\mathrm{tail}}(\mathcal{C}_0)$, and $D_\chi(\mathcal{C}) \le 0.95 D_\chi(\mathcal{C}_0)$ for every dominant $\chi$. Equation (57) then selects $\mathcal{C}_1$.

TABLE IX: The finite set $\mathcal{U}_0$ is constructed from the updates allowed at iteration 0.

| $\ell$ | $\mathcal{C}_{q,\ell}$ | update | update quantities | $\mathcal{U}_q^{\mathrm{eval}}$ |
|---|---|---|---|---|
| 1 | $\mathcal{C}_{0,1}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(1)})$ | 0 |
| 2 | $\mathcal{C}_{0,2}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(2)})$ | 1 |
| 3 | $\mathcal{C}_{0,3}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(3)})$ | 1 |
| 4 | $\mathcal{C}_{0,4}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(4)})$ | 1 |
| 5 | $\mathcal{C}_{0,5}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(1)})$ | 1 |
| 6 | $\mathcal{C}_{0,6}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(2)})$ | 1 |
| 7 | $\mathcal{C}_{0,7}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(3)})$ | 0 |
| 8 | $\mathcal{C}_{0,8}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(4)})$ | 1 |
| 9 | $\mathcal{C}_{0,9}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(1)})$ | 1 |
| 10 | $\mathcal{C}_{0,10}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(2)})$ | 0 |
| 11 | $\mathcal{C}_{0,11}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(3)})$ | 1 |
| 12 | $\mathcal{C}_{0,12}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(4)})$ | 0 |
| 13 | $\mathcal{C}_{0,13}$ | $U_{\mathrm{graph}}$ | $(33, 1)$ | 1 |
| 14 | $\mathcal{C}_{0,14}$ | $U_{\mathrm{inv}}$ | $(33, 1)$ | 1 |
| 15 | $\mathcal{C}_{0,15}$ | $U_{\mathrm{inv}}$ | $(33, 10)$ | 1 |

TABLE X: The candidates in $\mathcal{U}_0^{\mathrm{eval}}$ are tested by (56).

| $\ell$ | $D_{\mathrm{out}}$ | $D_{\mathrm{inv}}$ | $D_{\mathrm{graph}}$ | $D_{\mathrm{MC}}$ | $D_{\max}$ | $C_{\mathrm{tail}}$ | Comp | $\mathcal{A}_q^{\mathrm{rep}}$ | rank | selected |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1.0590 | 341.91 | 1.6656e+05 | 8.7219 | 1.6656e+05 | 13.2202 | 0.13223 | 0 | – | 0 |
| 3 | 0.93262 | 185.80 | 2.1004e+04 | 0.97067 | 2.1004e+04 | 1.67860 | 0.13223 | 0 | – | 0 |
| 4 | 1.0690 | 137.32 | 1.3211e+04 | 0.63726 | 1.3211e+04 | 1.05850 | 0.13223 | 0 | – | 0 |
| 5 | 0.94132 | 186.25 | 1.9338e+04 | 0.86931 | 1.9338e+04 | 1.54668 | 0.13223 | 0 | – | 0 |
| 6 | 1.0351 | 321.60 | 1.2721e+05 | 6.3195 | 1.2721e+05 | 10.1016 | 0.13223 | 0 | – | 0 |
| 8 | 1.0492 | 151.45 | 1.5928e+04 | 0.74079 | 1.5928e+04 | 1.27458 | 0.13223 | 0 | – | 0 |
| 9 | 0.92440 | 189.09 | 1.9271e+04 | 0.83688 | 1.9271e+04 | 1.54155 | 0.13223 | 0 | – | 0 |
| 11 | 0.90019 | 200.28 | 2.3616e+04 | 1.0267 | 2.3616e+04 | 1.88636 | 0.13223 | 0 | – | 0 |
| 13 | 1.0690 | 116.02 | 1.1450e+04 | 0.63728 | 1.1450e+04 | 0.915730 | 0.27273 | 1 | 1 | 1 |
| 14 | 1.0690 | 117.77 | 1.2171e+04 | 0.63723 | 1.2171e+04 | 0.974618 | 0.27273 | 1 | 2 | 0 |
| 15 | 1.0690 | 121.64 | 1.4679e+04 | 0.63722 | 1.4679e+04 | 1.17267 | 0.27273 | 0 | – | 0 |

At $q = 1$, the same three inequalities use $\mathcal{C}_1$ on their right hand sides and select $\mathcal{C}_2$.

TABLE XI: The finite set $\mathcal{U}_1$ is constructed from the updates allowed at iteration 1.

| $\ell$ | $\mathcal{C}_{q,\ell}$ | update | update quantities | $\mathcal{U}_q^{\mathrm{eval}}$ |
|---|---|---|---|---|
| 1 | $\mathcal{C}_{1,1}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(1)})$ | 0 |
| 2 | $\mathcal{C}_{1,2}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(2)})$ | 1 |
| 3 | $\mathcal{C}_{1,3}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(3)})$ | 1 |
| 4 | $\mathcal{C}_{1,4}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(4)})$ | 1 |
| 5 | $\mathcal{C}_{1,5}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(1)})$ | 1 |
| 6 | $\mathcal{C}_{1,6}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(2)})$ | 1 |
| 7 | $\mathcal{C}_{1,7}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(3)})$ | 0 |

| | | | | |
|---|---|---|---|---|
| 8 | $\mathcal{C}_{1,8}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(4)})$ | 1 |
| 9 | $\mathcal{C}_{1,9}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(1)})$ | 1 |
| 10 | $\mathcal{C}_{1,10}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(2)})$ | 0 |
| 11 | $\mathcal{C}_{1,11}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(3)})$ | 1 |
| 12 | $\mathcal{C}_{1,12}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(4)})$ | 0 |
| 13 | $\mathcal{C}_{1,13}$ | $U_{\mathrm{graph}}$ | $(49, 1)$ | 1 |
| 14 | $\mathcal{C}_{1,14}$ | $U_{\mathrm{inv}}$ | $(49, 1)$ | 1 |
| 15 | $\mathcal{C}_{1,15}$ | $U_{\mathrm{inv}}$ | $(49, 10)$ | 1 |

TABLE XII: The candidates in $\mathcal{U}_1^{\mathrm{eval}}$ are tested by (56).

| $\ell$ | $D_{\mathrm{out}}$ | $D_{\mathrm{inv}}$ | $D_{\mathrm{graph}}$ | $D_{\mathrm{MC}}$ | $D_{\max}$ | $C_{\mathrm{tail}}$ | Comp | $\mathcal{A}_q^{\mathrm{rep}}$ | rank | selected |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1.0719 | 269.33 | 1.4693e+05 | 8.7220 | 1.4693e+05 | 11.6566 | 0.27273 | 0 | – | 0 |
| 3 | 0.93951 | 147.22 | 1.8482e+04 | 0.97076 | 1.8482e+04 | 1.47401 | 0.27273 | 0 | – | 0 |
| 4 | 1.0746 | 108.77 | 1.1648e+04 | 0.63728 | 1.1648e+04 | 0.930812 | 0.27273 | 0 | – | 0 |
| 5 | 0.94904 | 147.64 | 1.7078e+04 | 0.86933 | 1.7078e+04 | 1.36281 | 0.27273 | 0 | – | 0 |
| 6 | 1.0479 | 253.51 | 1.1231e+05 | 6.3195 | 1.1231e+05 | 8.91319 | 0.27273 | 0 | – | 0 |
| 8 | 1.0565 | 119.90 | 1.4069e+04 | 0.74084 | 1.4069e+04 | 1.12314 | 0.27273 | 0 | – | 0 |
| 9 | 0.93219 | 149.94 | 1.7035e+04 | 0.83689 | 1.7035e+04 | 1.35947 | 0.27273 | 0 | – | 0 |
| 11 | 0.90785 | 158.80 | 2.0837e+04 | 1.0267 | 2.0837e+04 | 1.66107 | 0.27273 | 0 | – | 0 |
| 13 | 1.0746 | 0.0043698 | 0.51100 | 0.63738 | 1.0746 | 0.00393537 | 0.40496 | 1 | 2 | 0 |
| 14 | 1.0746 | 0.0043736 | 0.51273 | 0.63736 | 1.0746 | 0.00393544 | 0.40496 | 1 | 3 | 0 |
| 15 | 1.0746 | 0.0043731 | 0.51105 | 0.63737 | 1.0746 | 0.00393531 | 0.40496 | 1 | 1 | 1 |

At $q = 2$, the right hand sides use $\mathcal{C}_2$ and the selected $U_{\mathrm{out}}$ member defines $\mathcal{C}_3 = \mathcal{C}^\star$.

TABLE XIII: The finite set $\mathcal{U}_2$ is constructed from the updates allowed at iteration 2.

| $\ell$ | $\mathcal{C}_{q,\ell}$ | update | update quantities | $\mathcal{U}_q^{\mathrm{eval}}$ |
|---|---|---|---|---|
| 1 | $\mathcal{C}_{2,1}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(1)})$ | 0 |
| 2 | $\mathcal{C}_{2,2}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(2)})$ | 1 |
| 3 | $\mathcal{C}_{2,3}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(3)})$ | 1 |
| 4 | $\mathcal{C}_{2,4}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.2), \varpi^{(4)})$ | 1 |
| 5 | $\mathcal{C}_{2,5}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(1)})$ | 1 |
| 6 | $\mathcal{C}_{2,6}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(2)})$ | 1 |
| 7 | $\mathcal{C}_{2,7}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(3)})$ | 0 |
| 8 | $\mathcal{C}_{2,8}$ | $U_{\mathrm{out}}$ | $(\Gamma(0.6), \varpi^{(4)})$ | 1 |
| 9 | $\mathcal{C}_{2,9}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(1)})$ | 1 |
| 10 | $\mathcal{C}_{2,10}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(2)})$ | 0 |
| 11 | $\mathcal{C}_{2,11}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(3)})$ | 1 |
| 12 | $\mathcal{C}_{2,12}$ | $U_{\mathrm{out}}$ | $(\Gamma(1), \varpi^{(4)})$ | 0 |

TABLE XIV: The candidates in $\mathcal{U}_2^{\mathrm{eval}}$ are tested by (56).

| $\ell$ | $D_{\mathrm{out}}$ | $D_{\mathrm{inv}}$ | $D_{\mathrm{graph}}$ | $D_{\mathrm{MC}}$ | $D_{\max}$ | $C_{\mathrm{tail}}$ | Comp | $\mathcal{A}_q^{\mathrm{rep}}$ | rank | selected |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1.1038 | 0.011299 | 6.9840 | 8.7223 | 8.7223 | 0.00805556 | 0.40496 | 0 | – | 0 |
| 3 | 0.97073 | 0.0055981 | 0.77308 | 0.97091 | 0.97091 | 0.00375177 | 0.40496 | 1 | 2 | 0 |
| 4 | 1.1118 | 0.0043681 | 0.51098 | 0.63737 | 1.1118 | 0.00406069 | 0.40496 | 0 | – | 0 |
| 5 | 0.98003 | 0.0080711 | 0.91762 | 0.86948 | 0.98003 | 0.00374308 | 0.40496 | 1 | 3 | 0 |
| 6 | 1.0790 | 0.0099122 | 4.8215 | 6.3198 | 6.3198 | 0.00676562 | 0.40496 | 0 | – | 0 |
| 8 | 1.0919 | 0.0046230 | 0.59067 | 0.74092 | 1.0919 | 0.00404502 | 0.40496 | 0 | – | 0 |
| 9 | 0.96248 | 0.0053744 | 0.66677 | 0.83694 | 0.96248 | 0.00365731 | 0.40496 | 1 | 1 | 1 |
| 11 | 0.93733 | 0.0056591 | 0.80901 | 1.0269 | 1.0269 | 0.00366673 | 0.40496 | 1 | 4 | 0 |

For the selected candidates, define $m_q(w) = \|\widehat{\sigma}_q(w) - z_w^{(q)}\|_2$ for $w \in W$, where $z_w^{(q)}$ is the terminal state of the query initialized at $c_q$. Fig. 3 shows $m_q$ on the same grid of exosystem states and color scale. Fig. 3(a) shows the initial mismatch. Fig. 3(b) shows its reduction after the selected $U_{\mathrm{graph}}$ update. Fig. 3(c) shows the further reduction obtained by the selected $U_{\mathrm{inv}}$ update. Fig. 3(d) evaluates the recomputed frozen controller and graph after $U_{\mathrm{out}}$.

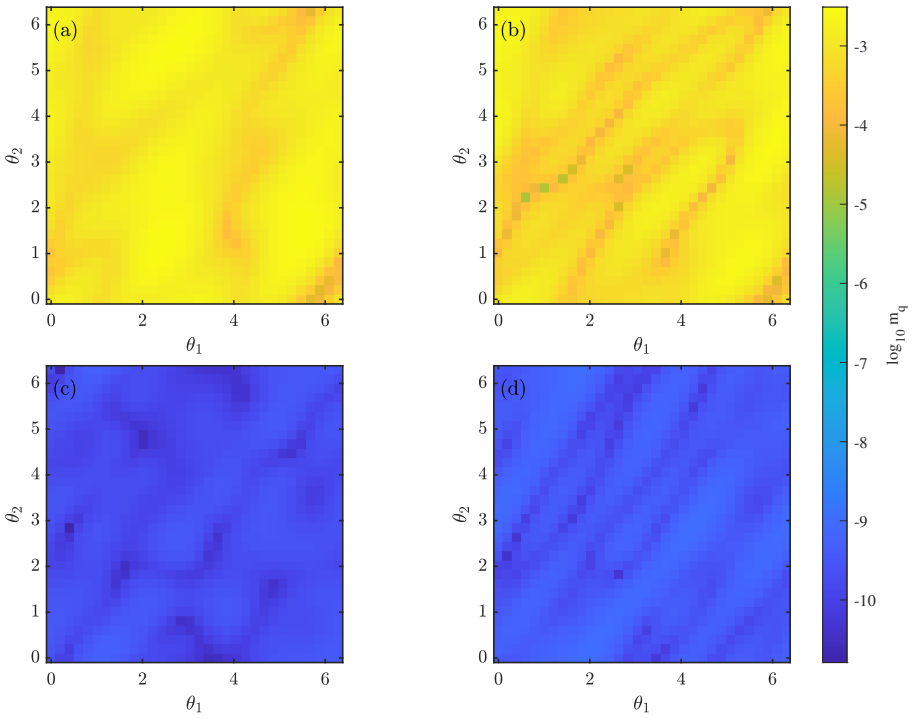



Fig. 3. The selected updates reduce $m_q(w)$ over the torus. (a) $\mathcal{C}_0$. (b) $\mathcal{C}_1$ after $U_{\mathrm{graph}}$. (c) $\mathcal{C}_2$ after $U_{\mathrm{inv}}$. (d) $\mathcal{C}^\star$ after $U_{\mathrm{out}}$.

The frozen deterministic bounds are $\bar{L}_F^{z,\star} \approx 124.13$, $\bar{Z}_a^\star \approx 5.6315$, $\bar{A}_K^\star \approx 104.98$, $\bar{m}_{\mathrm{qry}}^\star \approx 5.3090$, $\widehat{\delta}_z^\star \approx 5.595 \times 10^{-6}$, $\widehat{t}_\Theta^\star \approx 0.04296$, $\bar{R}_F^{\mathrm{comp},\star} \approx 606.90$, and $\bar{Y}_F^{\mathrm{comp},\star} \approx 1924.88$. The frozen matrices and centers are

TABLE XV
FROZEN METRIC MATRIX $M^\star$.

| | | | | | |
|---|---|---|---|---|---|
| 0.4779 | -0.1288 | 0.01794 | 0.1039 | 0.1343 | 0.05201 |
| -0.1288 | 0.6802 | -0.01537 | 0.2699 | 0.2766 | -0.03544 |
| 0.01794 | -0.01537 | 0.1426 | 0.001842 | -0.002664 | 0.02377 |
| 0.1039 | 0.2699 | 0.001842 | 0.3883 | 0.1334 | -0.01632 |
| 0.1343 | 0.2766 | -0.002664 | 0.1334 | 0.4062 | 0.02448 |
| 0.05201 | -0.03544 | 0.02377 | -0.01632 | 0.02448 | 0.06616 |

TABLE XVI: Frozen RKHS centers.

| $i$ | $w_1$ | $w_2$ | $w_3$ | $w_4$ |
|---|---|---|---|---|
| 1 | 0.03162 | -0.9995 | 0.2923 | -0.9563 |
| 2 | -0.9342 | 0.3567 | 0.9929 | -0.1192 |
| 3 | 0.9712 | -0.2384 | 0.7868 | 0.6172 |
| 4 | 0.9177 | 0.3972 | -0.06405 | 0.9979 |
| 5 | 0.5202 | -0.854 | -0.5876 | -0.8091 |
| 6 | -0.4106 | 0.9118 | -0.9876 | -0.1573 |
| 7 | -0.9598 | -0.2807 | 0.6599 | 0.7514 |
| 8 | -0.5949 | -0.8038 | -0.2465 | 0.9691 |
| 9 | 0.3995 | 0.9167 | 0.593 | -0.8052 |
| 10 | 0.9943 | -0.1068 | -0.8354 | 0.5496 |
| 11 | 0.6294 | -0.777 | 0.954 | -0.2998 |
| 12 | -0.8783 | 0.4781 | -0.4288 | -0.9034 |
| 13 | -0.9138 | -0.4061 | -0.9222 | 0.3866 |
| 14 | 0.5181 | 0.8553 | -0.952 | -0.3062 |
| 15 | -0.236 | 0.9717 | 0.9994 | 0.03371 |
| 16 | -0.1019 | -0.9948 | -0.9996 | 0.02702 |
| 17 | -0.7552 | 0.6555 | -0.2771 | 0.9608 |

| $i$ | $w_1$ | $w_2$ | $w_3$ | $w_4$ |
|---|---|---|---|---|
| 18 | -0.7536 | -0.6573 | 0.8266 | -0.5628 |
| 19 | 0.7274 | -0.6862 | -0.01921 | 0.9998 |
| 20 | 0.1116 | 0.9937 | -0.4205 | 0.9073 |
| 21 | -0.945 | 0.327 | 0.5672 | -0.8236 |
| 22 | -0.989 | 0.1478 | -0.6649 | 0.747 |
| 23 | 0.8962 | 0.4436 | -0.9909 | 0.135 |
| 24 | -0.07048 | -0.9975 | -0.6408 | 0.7677 |
| 25 | -0.2853 | 0.9584 | 0.165 | 0.9863 |
| 26 | -0.9977 | -0.06803 | 0.003345 | -1 |
| 27 | -0.7207 | 0.6932 | 0.8869 | 0.462 |
| 28 | -0.4392 | 0.8984 | -0.7711 | 0.6367 |
| 29 | -0.9999 | -0.01699 | -0.9795 | -0.2014 |
| 30 | -0.6218 | 0.7831 | 0.1559 | -0.9878 |
| 31 | 0.6518 | 0.7584 | -0.7417 | 0.6707 |
| 32 | -0.5968 | 0.8024 | 0.844 | -0.5363 |
| 33 | -0.8528 | 0.5223 | -0.9704 | 0.2413 |
| 34 | 0.9996 | 0.02675 | -0.2554 | -0.9668 |
| 35 | -0.3132 | -0.9497 | 0.7668 | 0.6419 |
| 36 | 0.9816 | 0.1909 | 0.709 | -0.7052 |
| 37 | -0.4084 | -0.9128 | -0.4573 | -0.8893 |
| 38 | 0.2455 | -0.9694 | 0.4144 | 0.9101 |
| 39 | -0.8345 | -0.551 | -0.7965 | -0.6047 |
| 40 | 0.9871 | 0.1598 | 0.9979 | 0.06541 |
| 41 | 0.8291 | 0.5591 | 0.1872 | -0.9823 |
| 42 | 0.9307 | -0.3658 | 0.3349 | -0.9422 |
| 43 | 0.276 | -0.9612 | 0.9567 | 0.291 |
| 44 | 0.4759 | -0.8795 | -0.9095 | 0.4157 |
| 45 | 0.9984 | -0.05588 | -0.7043 | -0.7099 |
| 46 | 0.5631 | -0.8264 | 0.6766 | -0.7363 |
| 47 | 0.9821 | -0.1885 | -0.4491 | 0.8935 |
| 48 | 0.4479 | -0.8941 | -0.9133 | -0.4073 |
| 49 | 0.9474 | 0.3201 | 0.4431 | 0.8965 |

TABLE XVII: Frozen RKHS coefficient matrix by center.

| $i$ | $\widehat{\Theta}_{1i}$ | $\widehat{\Theta}_{2i}$ | $\widehat{\Theta}_{3i}$ | $\widehat{\Theta}_{4i}$ | $\widehat{\Theta}_{5i}$ | $\widehat{\Theta}_{6i}$ |
|---|---|---|---|---|---|---|
| 1 | 5.150e-04 | 4.156e-04 | -3.459e-09 | -3.293e-04 | -4.053e-04 | -1.733e-05 |
| 2 | 1.866e-05 | -4.893e-05 | -3.637e-10 | 1.186e-05 | 1.715e-05 | -9.550e-06 |
| 3 | -2.949e-04 | -1.732e-04 | 3.998e-09 | 1.654e-04 | 2.012e-04 | 1.924e-05 |
| 4 | 0.001456 | 0.001124 | -1.152e-08 | -9.106e-04 | -0.001119 | -5.519e-05 |
| 5 | -1.044e-04 | -4.326e-05 | 1.089e-09 | 5.240e-05 | 6.296e-05 | 9.509e-06 |
| 6 | -1.208e-04 | -1.480e-04 | 1.623e-10 | 9.547e-05 | 1.192e-04 | -2.394e-06 |
| 7 | -2.003e-04 | -1.291e-04 | 1.542e-09 | 1.157e-04 | 1.410e-04 | 1.127e-05 |
| 8 | 4.314e-04 | 3.514e-04 | -3.584e-09 | -2.753e-04 | -3.386e-04 | -1.364e-05 |
| 9 | 4.607e-05 | 1.017e-05 | -2.336e-10 | -1.949e-05 | -2.277e-05 | -5.702e-06 |
| 10 | -5.390e-04 | -4.423e-04 | 5.297e-09 | 3.462e-04 | 4.264e-04 | 1.598e-05 |
| 11 | 5.347e-04 | 4.359e-04 | -2.528e-09 | -3.428e-04 | -4.219e-04 | -1.737e-05 |
| 12 | 2.237e-04 | 2.221e-04 | -4.871e-10 | -1.585e-04 | -1.967e-04 | -1.926e-06 |
| 13 | 9.550e-04 | 7.938e-04 | -7.004e-09 | -6.173e-04 | -7.604e-04 | -2.832e-05 |
| 14 | -3.211e-04 | -3.175e-04 | 2.833e-09 | 2.272e-04 | 2.821e-04 | 2.256e-06 |
| 15 | 4.992e-05 | -6.773e-05 | -7.035e-10 | 7.793e-06 | 1.400e-05 | -1.735e-05 |
| 16 | -5.859e-04 | -4.618e-04 | 4.017e-09 | 3.697e-04 | 4.545e-04 | 2.081e-05 |
| 17 | 5.298e-04 | 3.774e-04 | -3.630e-09 | -3.177e-04 | -3.888e-04 | -2.396e-05 |
| 18 | -1.532e-04 | -6.267e-05 | 8.157e-10 | 7.539e-05 | 9.026e-05 | 1.403e-05 |
| 19 | 0.001046 | 8.148e-04 | -6.874e-09 | -6.564e-04 | -8.063e-04 | -3.926e-05 |

| $i$ | $\widehat{\Theta}_{1i}$ | $\widehat{\Theta}_{2i}$ | $\widehat{\Theta}_{3i}$ | $\widehat{\Theta}_{4i}$ | $\widehat{\Theta}_{5i}$ | $\widehat{\Theta}_{6i}$ |
|---|---|---|---|---|---|---|
| 20 | -1.803e-04 | -1.717e-04 | 1.030e-09 | 1.251e-04 | 1.547e-04 | 2.971e-06 |
| 21 | 1.194e-04 | 1.273e-04 | 3.725e-11 | -8.892e-05 | -1.108e-04 | -3.562e-08 |
| 22 | -0.001189 | -9.114e-04 | 8.292e-09 | 7.393e-04 | 9.074e-04 | 4.625e-05 |
| 23 | 0.00115 | 9.325e-04 | -9.734e-09 | -7.360e-04 | -9.064e-04 | -3.659e-05 |
| 24 | -7.392e-04 | -5.332e-04 | 6.030e-09 | 4.476e-04 | 5.481e-04 | 3.339e-05 |
| 25 | -1.911e-04 | -1.789e-04 | 1.352e-09 | 1.292e-04 | 1.597e-04 | 1.990e-06 |
| 26 | -2.562e-04 | -2.283e-04 | 9.689e-10 | 1.713e-04 | 2.115e-04 | 5.851e-06 |
| 27 | -1.284e-04 | -5.527e-05 | 1.393e-09 | 6.355e-05 | 7.604e-05 | 1.134e-05 |
| 28 | -6.370e-05 | -1.997e-05 | 1.262e-10 | 2.732e-05 | 3.203e-05 | 6.168e-06 |
| 29 | -0.001146 | -9.435e-04 | 7.360e-09 | 7.370e-04 | 9.073e-04 | 3.571e-05 |
| 30 | -2.024e-04 | -2.756e-04 | 7.874e-10 | 1.715e-04 | 2.156e-04 | -8.632e-06 |
| 31 | -5.334e-04 | -4.463e-04 | 4.790e-09 | 3.457e-04 | 4.262e-04 | 1.462e-05 |
| 32 | -1.419e-04 | -4.650e-05 | 1.440e-09 | 6.536e-05 | 7.767e-05 | 1.437e-05 |
| 33 | 7.671e-04 | 5.902e-04 | -4.371e-09 | -4.770e-04 | -5.853e-04 | -2.975e-05 |
| 34 | 3.227e-04 | 3.516e-04 | -4.237e-09 | -2.395e-04 | -2.984e-04 | 2.475e-06 |
| 35 | 1.594e-04 | 1.815e-04 | -1.688e-09 | -1.202e-04 | -1.500e-04 | 2.644e-06 |
| 36 | -7.972e-04 | -5.570e-04 | 6.387e-09 | 4.778e-04 | 5.849e-04 | 3.855e-05 |
| 37 | -4.173e-04 | -3.010e-04 | 2.758e-09 | 2.525e-04 | 3.091e-04 | 1.895e-05 |
| 38 | -4.656e-04 | -3.341e-04 | 4.813e-09 | 2.816e-04 | 3.450e-04 | 2.115e-05 |
| 39 | 6.600e-04 | 5.660e-04 | -4.593e-09 | -4.337e-04 | -5.350e-04 | -1.725e-05 |
| 40 | 7.709e-04 | 5.245e-04 | -5.493e-09 | -4.555e-04 | -5.565e-04 | -3.967e-05 |
| 41 | 2.661e-04 | 1.037e-04 | -1.710e-09 | -1.276e-04 | -1.524e-04 | -2.412e-05 |
| 42 | 6.803e-04 | 4.469e-04 | -2.946e-09 | -3.968e-04 | -4.841e-04 | -3.737e-05 |
| 43 | 5.012e-05 | 3.008e-05 | 3.645e-11 | -2.745e-05 | -3.314e-05 | -3.242e-06 |
| 44 | 8.755e-04 | 7.230e-04 | -5.416e-09 | -5.645e-04 | -6.953e-04 | -2.662e-05 |
| 45 | -6.344e-04 | -5.622e-04 | 7.117e-09 | 4.236e-04 | 5.233e-04 | 1.361e-05 |
| 46 | -8.880e-04 | -6.374e-04 | 5.788e-09 | 5.392e-04 | 6.605e-04 | 4.136e-05 |
| 47 | -8.761e-04 | -6.582e-04 | 7.817e-09 | 5.407e-04 | 6.631e-04 | 3.630e-05 |
| 48 | 3.957e-04 | 3.270e-04 | -2.336e-09 | -2.566e-04 | -3.164e-04 | -1.216e-05 |
| 49 | -8.544e-04 | -6.947e-04 | 7.236e-09 | 5.459e-04 | 6.721e-04 | 2.653e-05 |

Table XVIII lists the initial augmented states $z_0^{(\ell)}$ and exosystem states $w_0^{(\ell)}$ used in Fig. 1.

TABLE XVIII
THE FIVE INITIAL CONDITIONS GENERATE THE TRAJECTORIES IN FIG. 1.

| $\ell$ | $z_1$ | $z_2$ | $z_3$ | $z_4$ | $z_5$ | $z_6$ | $w_1$ | $w_2$ | $w_3$ | $w_4$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.0002 | 0.0403 | -0.0819 | 0.0655 | 0.0316 | -0.0858 | 0.2309 | -0.9730 | 0.9906 | -0.1365 |
| 2 | 0.1029 | 0.3028 | 0.1973 | -0.1257 | -0.2477 | 0.1679 | 0.3281 | -0.9446 | -0.9866 | -0.1632 |
| 3 | -0.1752 | 0.2973 | 0.0081 | -0.2625 | -0.0177 | -0.1775 | 0.3918 | -0.9201 | 0.1124 | -0.9937 |
| 4 | 0.4664 | 0.1486 | -0.6086 | -0.0449 | 0.0822 | 0.3975 | -0.9979 | -0.0644 | -0.9848 | -0.1737 |
| 5 | 0.4584 | -0.3312 | -0.0914 | 0.1458 | 0.2744 | 0.0752 | 0.9959 | 0.0905 | 0.5615 | -0.8275 |

## IV. PAC GUARANTEES

The final parameters are $N = 30$, $\gamma = 0.1$, $k_{\mathrm{disc}}^{\mathrm{quant}} = k_{\mathrm{disc}}^{\mathrm{ord}} = 0$, $\epsilon_{\mathrm{req}} = 0.004$, $\epsilon_{\mathrm{MC}}^{\mathrm{val}} = 2.5 \times 10^{-4}$, $\beta_{\mathrm{MC}}^{\mathrm{val}} = 0.02$, and $\delta_{\mathrm{quant}} = \delta_{\mathrm{ord}} = 0.05$. The distribution $\mu$ is implemented by sampling $\theta_1$ and $\theta_2$ independently and uniformly on $[0, 2\pi)$ and setting $w = (\cos\theta_1, \sin\theta_1, \cos\theta_2, \sin\theta_2)$. Before generating $\mathcal{D}_{\mathrm{val}}$, the frozen quantities give $a^\star = L_h\kappa_{M^\star}/\lambda^\star \approx 4.2164$, $\eta_{N_{\mathrm{lb}}^{\mathrm{val}}} \approx 5.9292 \times 10^{-5}$, and $a^\star\eta_{N_{\mathrm{lb}}^{\mathrm{val}}} \le \epsilon_{\mathrm{MC}}^{\mathrm{val}} = 2.5 \times 10^{-4}$. Lemma 6 gives $N_{\mathrm{lb}}^{\mathrm{val}} = 3{,}853{,}873{,}051{,}311{,}706$. Write $w = (\cos\theta_1, \sin\theta_1, \cos\theta_2, \sin\theta_2)$, and divide $[0, 2\pi)^2$ into $1024^2$ subsets of side length $2\pi/1024$. For the center $\theta^c$ of each subset, the following bound is applied to every entry of $R_F$:

$$|[R_F(\theta)]_i| \le |[R_F(\theta^c)]_i| + \frac{\pi}{1024}\sum_{a=1}^{2}\left|\frac{\partial[R_F]_i}{\partial\theta_a}(\theta^c)\right| + \frac{1}{2}\left(\frac{\pi}{1024}\right)^2 \sup_{\xi}\sum_{a,b=1}^{2}\left|\frac{\partial^2[R_F]_i}{\partial\theta_a\partial\theta_b}(\xi)\right|,$$

where the supremum is taken over that subset. On every subset, the values and the first and second derivatives of $R_F$ are evaluated by interval propagation with outward rounding. For this example, the exact vector field is used to evaluate the bounds in Remark 8 of the main paper. Taking the Euclidean norm of the resulting entrywise bounds and then the maximum over the $1024^2$ subsets gives

$$\sup_{w\in W}\|R_F(w)\|_2 \le 1.0227\times 10^{-7}.$$

With $T_{\mathrm{qry}}=30$, the contraction inequality and Lemma 2 give

$$m(w)\le \frac{e^{-\lambda^\star T_{\mathrm{qry}}}(R^\star-\epsilon_{\mathrm{safe}})+\sqrt{\lambda_{\max}(M^\star)}\dfrac{1-e^{-\lambda^\star T_{\mathrm{qry}}}}{\lambda^\star}\sup_{\xi\in W}\|R_F(\xi)\|_2}{\sqrt{\lambda_{\min}(M^\star)}}.$$

Substituting the frozen quantities gives

$$m(w)\le 2.4894\times 10^{-7}.$$

Consequently, using the frozen bound $\overline{L}_F^z$,

$$\begin{aligned}B_F(w)&\le \sup_{\xi\in W}\|R_F(\xi)\|_2+2\overline{L}_F^z m(w)\\&\le 6.1903\times 10^{-5}.\end{aligned}$$

The partition in Remark 8 of the main paper has $L=8193$, with $\mathcal{L}_\ell=[30(\ell-1)/8192,30\ell/8192)$ for $\ell=1,\dots,8192$ and $\mathcal{L}_{8193}=[30,\infty)$. The bounds are $b_{j,\ell}=6.1903\times 10^{-5}$ for $\ell=1,\dots,8192$ and $b_{j,8193}=\overline{Y}_F^{\mathrm{comp},\star}$. The corresponding probabilities are $p_\ell=\exp(-30(\ell-1)/8192)-\exp(-30\ell/8192)$ for $\ell=1,\dots,8192$ and $p_{8193}=\exp(-30)$. The 30 primary exosystem states require 30 residual queries. The $30N_{\mathrm{lb}}^{\mathrm{val}}\approx 1.156\times 10^{17}$ lookback variables are represented by 30 count vectors with 8193 entries. For each $j$, the corresponding count vector gives the upper bound on $\overline{Y}_j$ in Remark 8 of the main paper. For each $j$, define

$$\overline{Y}_j=\frac{1}{N_{\mathrm{lb}}^{\mathrm{val}}}\sum_{k=1}^{N_{\mathrm{lb}}^{\mathrm{val}}}B_F(w_{j,k}^-).$$

In Table XIX, the column headed $\overline{Y}_j$ contains this upper bound. The table also lists $B_{e,j}$, $S_{\lambda,j}^{\mathrm{MC}}$, $\mathbf{1}_{\{S_{\lambda,j}^{\mathrm{MC}}>\epsilon_{\mathrm{req}}\}}$, and the rank of $S_{\lambda,j}^{\mathrm{MC}}$ in the ordered sample.

TABLE XIX: Validation quantities for Theorems 2 and 3.

| $j$ | $w_1$ | $w_2$ | $w_3$ | $w_4$ | $B_{e,j}$ | $\overline{Y}_j$ | $S_{\lambda,j}^{\mathrm{MC}}$ | $\mathbf{1}_{\{S_{\lambda,j}^{\mathrm{MC}}>\epsilon_{\mathrm{req}}\}}$ | rank |
|---|---|---|---|---|---|---|---|---|---|
| 1 | -1.0000 | -0.0026681 | 0.85346 | 0.52116 | 0.00235272 | 6.19031e-05 | 0.00286373 | 0 | 19 |
| 2 | 0.33927 | 0.94069 | 0.67812 | -0.73495 | 0.00107266 | 6.19031e-05 | 0.00158367 | 0 | 13 |
| 3 | -0.83361 | -0.55235 | 0.96484 | 0.26285 | 0.000975380 | 6.19031e-05 | 0.00148639 | 0 | 10 |
| 4 | 0.87140 | 0.49057 | -0.56167 | -0.82736 | 0.000615922 | 6.19031e-05 | 0.00112693 | 0 | 6 |
| 5 | -0.93180 | 0.36297 | -0.57078 | 0.82110 | 0.00247487 | 6.19031e-05 | 0.00298588 | 0 | 22 |
| 6 | -0.95101 | 0.30917 | -0.99883 | 0.048338 | 0.00238954 | 6.19031e-05 | 0.00290056 | 0 | 20 |
| 7 | -0.84296 | -0.53797 | -0.61645 | 0.78740 | 0.000331211 | 6.19031e-05 | 0.000842222 | 0 | 4 |
| 8 | 0.67931 | -0.73385 | 0.17650 | 0.98430 | 0.00348137 | 6.19031e-05 | 0.00399238 | 0 | 30 |
| 9 | 0.80331 | 0.59556 | -0.99903 | 0.044078 | 0.000416036 | 6.19031e-05 | 0.000927047 | 0 | 5 |
| 10 | -0.68709 | -0.72657 | -0.99904 | -0.043740 | 0.000163308 | 6.19031e-05 | 0.000674319 | 0 | 2 |
| 11 | -0.27302 | 0.96201 | 0.98046 | -0.19671 | 0.00251726 | 6.19031e-05 | 0.00302827 | 0 | 24 |
| 12 | 0.94287 | -0.33316 | 0.21846 | -0.97585 | 0.00227596 | 6.19031e-05 | 0.00278697 | 0 | 18 |
| 13 | 0.87953 | -0.47584 | -0.68869 | 0.72506 | 0.00312127 | 6.19031e-05 | 0.00363228 | 0 | 29 |
| 14 | -0.40419 | -0.91467 | 0.28597 | 0.95824 | 0.00129814 | 6.19031e-05 | 0.00180915 | 0 | 14 |
| 15 | -0.17871 | -0.98390 | 0.67590 | -0.73700 | 0.000986032 | 6.19031e-05 | 0.00149704 | 0 | 11 |
| 16 | 0.42710 | 0.90421 | -0.73950 | 0.67316 | 0.000814970 | 6.19031e-05 | 0.00132598 | 0 | 8 |
| 17 | 0.71005 | 0.70416 | 0.88924 | -0.45744 | 0.000293288 | 6.19031e-05 | 0.000804299 | 0 | 3 |
| 18 | -0.96938 | 0.24556 | 0.96526 | -0.26130 | 0.00293774 | 6.19031e-05 | 0.00344875 | 0 | 28 |
| 19 | -0.87515 | -0.48384 | -0.81337 | -0.58175 | 0.000942100 | 6.19031e-05 | 0.00145311 | 0 | 9 |
| 20 | 0.99990 | -0.014414 | -0.96110 | -0.27619 | 0.00189232 | 6.19031e-05 | 0.00240333 | 0 | 15 |
| 21 | -0.30165 | 0.95342 | -0.63100 | -0.77578 | 0.00249451 | 6.19031e-05 | 0.00300552 | 0 | 23 |
| 22 | -0.89615 | -0.44375 | -0.22949 | 0.97331 | 0.000707187 | 6.19031e-05 | 0.00121820 | 0 | 7 |
| 23 | 0.96655 | -0.25650 | 0.85205 | -0.52346 | 0.00254145 | 6.19031e-05 | 0.00305247 | 0 | 25 |
| 24 | 0.051831 | -0.99866 | 0.10801 | 0.99415 | 0.00259613 | 6.19031e-05 | 0.00310714 | 0 | 26 |
| 25 | 0.98800 | -0.15446 | 0.11105 | -0.99382 | 0.00204528 | 6.19031e-05 | 0.00255629 | 0 | 16 |
| 26 | -0.23410 | 0.97221 | 0.87764 | 0.47932 | 0.00239953 | 6.19031e-05 | 0.00291054 | 0 | 21 |
| 27 | 0.73780 | -0.67502 | -0.98678 | -0.16206 | 0.00272468 | 6.19031e-05 | 0.00323569 | 0 | 27 |
| 28 | 0.66386 | 0.74786 | 0.82686 | -0.56242 | 9.12237e-05 | 6.19031e-05 | 0.000602234 | 0 | 1 |
| 29 | -0.52659 | -0.85012 | -0.58055 | 0.81423 | 0.00101975 | 6.19031e-05 | 0.00153076 | 0 | 12 |
| 30 | 0.99341 | -0.11458 | -0.98043 | -0.19689 | 0.00210380 | 6.19031e-05 | 0.00261481 | 0 | 17 |